



Some (potentially) helpful formulas:

Thermodynamics:

$$dE = dQ + dW$$

$$C_X = \left. \frac{dQ}{dT} \right|_X$$

$$dE = TdS + \sum_i J_i dx_i + \mu dN$$

$$F = E - TS$$

$$G = E - TS - Jx$$

$$\Phi = E - TS - \mu N$$

Kinetic Theory of Gases

$$\frac{d\vec{q}_i}{dt} = \frac{\partial H}{\partial \vec{p}_i}$$

$$\frac{d\vec{p}_i}{dt} = -\frac{\partial H}{\partial \vec{q}_i}$$

$$\{A, B\} = \sum_{\alpha=1}^{3N} \left(\frac{\partial A}{\partial q_\alpha} \cdot \frac{\partial B}{\partial p_\alpha} - \frac{\partial A}{\partial p_\alpha} \cdot \frac{\partial B}{\partial q_\alpha} \right)$$

$$\frac{\partial \rho}{\partial t} = \{H, \rho\}$$

Student #6

Probability/Mathematical Formulas:

$$\tilde{p}(k) = \langle e^{-ikx} \rangle = \sum_{n=0}^{\infty} \frac{(-ik)^n}{n!} \langle x^n \rangle$$

$$\ln \tilde{p}(k) = \sum_{n=1}^{\infty} \frac{(-ik)^n}{n!} \langle x^n \rangle_c$$

$$\int_0^{\infty} dx x^n e^{-\alpha x} = \frac{n!}{\alpha^{n+1}}$$

$$\int_{-\infty}^{\infty} dx e^{-ikx - \frac{x^2}{2\sigma^2}} = \sqrt{2\pi\sigma^2} e^{-\frac{\sigma^2 k^2}{2}}$$

$$\int_{-\infty}^{\infty} x^2 e^{-ax^2} = \frac{1}{2} \sqrt{\frac{\pi}{a^3}}$$

$$\lim_{N \rightarrow \infty} N! = N \ln N - N$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n!}$$

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$$

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$$

$$\sinh x = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$$

$$\cosh x = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!}$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$$

Classical Statistical Mechanics

$$p_{(T,x,N)}(\mu_s) = \frac{e^{-\beta H(\mu_s)}}{Z(T,x,N)}$$

$$Z(T,x,N) = \sum_{\mu_s} e^{-\beta H(\mu_s)}$$

$$p_{(T,J,N)}(\mu_s) = \frac{e^{\beta \cdot J \cdot x - \beta H(\mu_s)}}{Z(T,J,N)}$$

$$Z(T,J,N) = \sum_{\mu_s} e^{-\beta(H(\mu_s) - Jx)}$$

$$p_{(T,\mu,x)}(\mu_s) = \frac{e^{\beta \cdot J \cdot x - \beta H(\mu_s)}}{Z(T,J,N)}$$

$$Q(T,\mu,x) = \sum_{\mu_s} e^{-\beta(H(\mu_s) - \mu N)}$$

$$S = k \ln \Omega$$

$$F = -k_B T \ln Z(T,x,N)$$

$$G = -k_B T \ln Z(T,J,N)$$

$$\Phi = -k_B T \ln Q(T,\mu,x)$$

$$\langle E \rangle = -\frac{\partial \ln Z}{\partial \beta}$$

$$\langle x \rangle = k_B T \frac{\partial \ln Z}{\partial J}$$

$$\langle N \rangle = \frac{\partial \ln Q}{\partial (\beta \mu)}$$

$$\lambda(T) = \frac{h}{\sqrt{2\pi m k_B T}}$$

Interacting Particles

$$\ln Z = \ln Z_0 + \frac{N(N-1)}{2V} \int d^3\vec{q} [\exp(-\beta V(\vec{q})) - 1] + O\left(\frac{N^3}{V^3}\right)$$

$$\frac{P}{k_B T} = \frac{N}{V} \left[1 + B_2(T) \frac{N}{V} + B_3(T) \frac{N^2}{V^2} + \dots \right]$$

$$f(\vec{q}) = \exp(-\beta V(\vec{q})) - 1$$

$$B_l(T) = -\frac{l-1}{l!} \bar{d}_l$$

$$(P - an^2)(V - Nb) = Nk_B T$$

$$Z(T, N, V) \approx \frac{(V - N\Omega/2)^2}{N! \lambda^{3N}} \exp \frac{\beta \mu N^2}{2V}$$

Quantum Statistical Mechanics

$$\eta = \begin{cases} +1 & \text{for bosons} \\ -1 & \text{for fermions} \end{cases}$$

$$\ln Q_\eta = -\eta \sum_{\vec{k}} \ln \left[1 - \eta \exp \left(\beta \mu - \beta \epsilon(\vec{k}) \right) \right]^{-1}$$

$$z = e^{\beta \mu}$$

$$N_\eta = \sum_{\vec{k}} \frac{1}{z^{-1} e^{\beta \epsilon(\vec{k})} - \eta}$$

$$E_\eta = \sum_{\vec{k}} \frac{\epsilon(\vec{k})}{z^{-1} e^{\beta \epsilon(\vec{k})} - \eta}$$

$$\beta P_\eta = \frac{g}{\lambda^3} f_{5/2}^\eta(z)$$

$$n_\eta = \frac{g}{\lambda^3} f_{3/2}^\eta(z)$$

$$\epsilon_\eta = \frac{3}{2} P_\eta$$

$$f_m^\eta(z) = \frac{1}{(m-1)!} \int_0^\infty \frac{dx}{x^{m-1}} z^{-1} e^x - \eta = \sum_{\alpha=1}^\infty \eta^{\alpha+1} \frac{z^\alpha}{\alpha^m}$$

$$\frac{d}{dz} f_m^\eta(z) = \frac{1}{z} f_{m-1}^\eta(z)$$

Some μ_s for 1)

All standing $\rightarrow H = W_e$

All down $\rightarrow H = 0$

Some standing $\rightarrow H = U_e$

23 Problem 1 (35 points)

Consider a polymer formed by connecting N disc-shaped molecules into a one-dimensional chain. Each molecule can align along either its long axis (of length $2a$) or short axis (of length a). Furthermore, when laying on its long axis there are two configuration it may assume, whereas when lying along its short axis there is only one possible configuration. The energy of the monomer aligned along its shorter axis is higher by ϵ , that is, the total energy of the system is $H = \epsilon U$ where U is the number of monomers standing up.

a (10 points) Calculate the partition function $Z(T, N)$ of the polymer.

side view L_x  top view  L_y

$$Z = \sum_{\mu_s} e^{-\beta H_{\mu_s}} \quad H = 0 \text{ when down (x2 because two configurations when down)}$$

$$N = U + N_{\downarrow}, \quad N_{\downarrow} = N - U$$

$$1 + e^{-\beta \epsilon U} \quad \text{Is it just } Z = 1 + e^{-\beta \epsilon U} \text{ ? No, we need } N \text{ dependence}$$

How many μ_s ? $\binom{N}{U} = \frac{N!}{U!(N-U)!}$

$$Z = \sum_{U=0}^N \left(e^{-\beta \epsilon U} \right) \left(\frac{N!}{U!(N-U)!} \right)$$

b (15 points) Calculate the average length $\langle L(T, N) \rangle$ and the variance of the length $\langle L(T, N)^2 \rangle_c$ of the polymer.

10 $\langle L \rangle = kT \frac{\partial \ln Z}{\partial p}$? Not this one?

$$\langle L \rangle = \sum L p(\mu_s) \quad \text{and} \quad p(\mu_s) = \frac{e^{-\beta E_{\mu_s}}}{Z} \Rightarrow p(U) = \frac{1}{N!} \frac{e^{-\beta \epsilon U}}{\sum_{U=0}^N e^{-\beta \epsilon U} \left(\frac{1}{U!(N-U)!} \right)}$$

$$\langle L \rangle = \sum L(T, N) \frac{1}{N!} \frac{e^{-\beta \epsilon U}}{\sum_{U=0}^N e^{-\beta \epsilon U} \left(\frac{1}{U!(N-U)!} \right)} \quad ?$$

variance B just $\langle L^2 \rangle_c = \langle L^2 \rangle - \langle L \rangle^2 \quad \checkmark$

c (10 points) Determine the energy per monomer and heat capacity per monomer. Sketch the heat capacity as a function of temperature.

are we looking for $\langle E \rangle$?

$$\langle E \rangle = -\frac{\partial \ln Z}{\partial \beta}$$

~~$$\ln Z = \sum_{U=0}^N \ln \left(\frac{N!}{U!(N-U)!} e^{-\beta \epsilon U} \right)$$~~

$$\Rightarrow \sum \left(\ln N! + \ln e^{-\beta \epsilon U} - \ln U! - \ln (N-U)! \right)$$

$$\langle E \rangle = \frac{\partial}{\partial \beta} (-\beta \epsilon U) = \epsilon U$$

when taking $\frac{\partial}{\partial \beta}$

This doesn't make sense... should depend on T it's wrong a.

$$C_v = \frac{d\langle E \rangle}{dT} \Big|_V$$

Above answer must be wrong !!

Problem 2 (35 points)

The pressure P of a gas is related to its density $n = N/V$ and temperature T by the truncated expansion:

$$P = k_B T n - \frac{b}{2} n^2 + \frac{c}{6} n^3$$

where b and c are assumed to be positive temperature-independent constants.

- a (15 points) Locate the critical temperature T_c below which the equation must be invalid, and the corresponding density n_c and pressure P_c of the critical point. Find the ratio $k_B T_c n_c / P_c$

$$P = kT \frac{N}{V} - \frac{b}{2} \left(\frac{N}{V} \right)^2 + \frac{c}{6} \left(\frac{N}{V} \right)^3$$

Pressure shouldn't be ~~negative~~ so

~~Answer~~

$$0 \leq P \Rightarrow 0 \leq kTn - \frac{b}{2}n^2 + \frac{c}{6}n^3 \Rightarrow \frac{b}{2}n^2 - \frac{c}{6}n^3 \leq kTn$$

$$\Rightarrow \frac{b}{2k}n - \frac{c}{6k}n^2 \leq T_c \Rightarrow \boxed{\frac{n}{2k} \left(b - \frac{c}{3}n \right) \leq T_c} \quad \text{Plug into } P$$

$$\text{Then } P \leq \frac{n^2}{2} \left(b - \frac{c}{3}n \right) - \frac{b}{2}n^2 + \frac{c}{6}n^3 \Rightarrow 0 = b \leq b$$

I don't remember how to get n_c . With n_c and T_c , then plug in to P to get P_c .

$$P = \frac{n^2}{2} \left(b - \frac{c}{3}n \right) - \frac{b}{2}n^2 + \frac{c}{6}n^3 \Rightarrow 0? \quad \text{That's not right}$$

ugh... $\hat{n}$

$$0 \leq P \Rightarrow 0 \leq kTn - \frac{b}{2}n^2 + \frac{c}{6}n^3 \Rightarrow 0 \leq kT - \frac{b}{2}n + \frac{c}{6}n^2$$

$$\Rightarrow \frac{b}{2}n - \frac{c}{6}n^2 \leq kT$$

b (10 points) Calculate the isothermal compressibility $\kappa_T = -\frac{1}{V} \frac{\partial V}{\partial P} \big|_T$ and sketch its behavior as a function of T for $n = n_c$.

2 $\frac{\partial V}{\partial P}$ Need formula with n_c . Then sub
in $\frac{N}{V}$ for n_c and solve for $V(P)$;

Then we can take derivative

c (10 points) On the critical isotherm give an expression for $(P - P_c)$ as a function of $(n - n_c)$

0 with expression for P_c and n_c

we can solve for this function at T_c

Problem 3 (30 points + 10 bonus)

Molecular hydrogen exists in two states: ortho and para.

a (20 points) In para-hydrogen, the two electrons of H_2 form a singlet (antisymmetric) state. Therefore, the orbital angular momentum can have only even values. That is, the Hamiltonian has the form:

2G

$$H = \frac{\hbar^2}{2I} l(l+1)$$

$2l+1$ total states

for $l = 0, 2, 4, \dots \Rightarrow l \rightarrow 2l$

1C

In contrast, in ortho-hydrogen, the electrons are in a triply degenerate symmetric state. Therefore, the orbital angular momentum can have only odd values. That is, the Hamiltonian has the form:

$$H = \frac{\hbar^2}{2I} l(l+1)$$

$3 \times (2l+1)$ total states

for $l = 1, 3, 5, \dots \Rightarrow l \rightarrow 2l+1$

Compute the rotational partition function for both para-hydrogen and ortho-hydrogen, and evaluate both of their low and high temperature limits.

Para $H = \frac{\hbar^2}{2I} (2l)(2l+1) = \frac{\hbar^2}{I} (2l^2 + l)$

$$Z_{\text{para}} = \sum_{l=0}^{2l+1} e^{-\beta H} = \sum e^{-\beta \frac{\hbar^2}{I} l(2l+1)} = 1 + e^{-\beta \frac{3\hbar^2}{I}} + e^{-\beta \frac{10\hbar^2}{I}} + e^{-\beta \frac{21\hbar^2}{I}} + \dots$$

Ortho $H = \frac{\hbar^2}{2I} (2l+1)(\overbrace{2l+1+1}^{2l+2}) = \frac{\hbar^2}{2I} (4l^2 + 6l + 2) = \frac{\hbar^2}{I} (2l^2 + 3l + 1)$

$$Z_{\text{ortho}} = 3 \sum_{l=0}^{2l+1} e^{-\beta H} = 3 \sum e^{-\beta \frac{\hbar^2}{I} (2l^2 + 3l + 1)} = 3e^{-\beta \frac{\hbar^2}{I}} + 3e^{-\beta \frac{6\hbar^2}{I}} + 3e^{-\beta \frac{15\hbar^2}{I}} + \dots$$

At low T , only first few l are available

$$Z_{\text{para}} \sim 1 + e^{-\beta \frac{3\hbar^2}{I}} \quad Z_{\text{ortho}} \sim 3e^{-\beta \frac{\hbar^2}{I}} \quad \text{of course!}$$

At high T , $\beta = \frac{1}{kT} \rightarrow 0$ Taylor expand? $\approx 1 - \beta \frac{\hbar^2}{I} l(2l+1)$ para

$$\approx 3 - 3\beta \frac{\hbar^2}{I} (2l^2 + 3l + 1) \text{ ortho}$$

b (10 points) For an equilibrium gas of N hydrogen molecules composed of N_p para and $N_o = N - N_p$ ortho molecules, calculate the partition function of the complete system. (you may ignore vibrational degrees of freedom)

Q

$$Z = \frac{(z_{\text{para}})^{N_p}}{N_p!} \cdot \frac{(z_{\text{ortho}})^{N_o}}{N_o!} = \frac{(z_p)^{N_p}}{N_p!} + \frac{(z_o)^{N-N_p}}{(N-N_p)!}$$

Factor how to get in terms of N ?

I remember this HW problem, but not how we did this :)

I remember it goes like $Z = \frac{1}{N!} (z_p + z_o)^N$ but I don't remember how we get there

c (10 points) Compute an expression for the rotational contribution to the internal energy $\langle E_{\text{rot}} \rangle$. Determine its behavior at the high and low temperature limits. Note that due to the large transition times between ortho- and para-hydrogen, in most circumstances the mixture is not actually in equilibrium.

Q

$$\langle E \rangle = - \frac{\partial \ln Z}{\partial \beta} \quad \text{so} \quad \ln Z = \ln \left(\frac{1}{N!} (z_p + z_o)^N \right) = N \ln(z_p + z_o) - \ln N!$$

$$\text{stirling} \approx N \ln(z_p + z_o) - \cancel{N \ln N} - \cancel{N}$$

when taking $\frac{\partial}{\partial \beta}$

chain rule

$$\langle E \rangle \approx -N \left(\frac{1}{z_p + z_o} \right) \frac{\partial}{\partial \beta} (z_p + z_o)$$

$$\text{low } T \approx \left(\frac{-N}{1 + e^{-\beta \frac{3\hbar^2}{I}} + 3e^{-\beta \frac{\hbar^2}{I}}} \right) \left(\frac{-3\hbar^2}{I} e^{-\beta \frac{3\hbar^2}{I}} - 3 \frac{\hbar^2}{I} e^{-\beta \frac{\hbar^2}{I}} \right)$$

$$\approx \frac{-N 3\hbar^2 (e^{-\beta \frac{3\hbar^2}{I}} + e^{-\beta \frac{\hbar^2}{I}})}{I (1 + e^{-\beta \frac{3\hbar^2}{I}} + 3e^{-\beta \frac{\hbar^2}{I}})} \quad ?$$